

Отчет по практической работе №1

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Группа: [БСБО-16-23]

Дата выполнения: [29.04.2026]

1. Выполненные команды Docker

1.1 Работа с образами

1.1 Работа с образами

```
PS D:\Dev\containerisation\containers-lab-1> docker search nginx
NAME                DESCRIPTION                                STARS     OFFICIAL
nginx               Official build of Nginx.                  21258     [OK]
nginx/nginx-ingress NGINX and NGINX Plus Ingress Controllers fo... 120
nginx/nginx-prometheus-exporter NGINX Prometheus Exporter for NGINX and NGIN... 51
nginx/unit          This repository is retired, use the Docker o... 66
nginx/nginx-ingress-operator NGINX Ingress Operator for NGINX and NGINX P... 3
nginx/nginx-quic-qns NGINX QUIC interop                        1
nginx/nginxaas-loadbalancer-kubernetes          1
nginx/unit-preview  Unit preview features                    0
bitnami/nginx       Bitnami Secure Image for nginx           203
bitnamicharts/nginx Bitnami Helm chart for NGINX Open Source     3
ubuntu/nginx        Nginx, a high-performance reverse proxy & we... 141

PS D:\Dev\containerisation\containers-lab-1> docker pull nginx:alpine
alpine: Pulling from library/nginx
6a0ac1617861: Downloading 786.4kB/3.864MB
82736a35d0e7: Downloading 852kB/1.871MB
583599bb7d38: Download complete
aee4e54b3865: Download complete
781ff50d2644: Waiting
453da7dbc73e: Waiting
4a8b0b2a5b19: Waiting
612c0c1df4c5: Waiting

PS D:\Dev\containerisation\containers-lab-1> docker images
REPOSITORY          TAG          IMAGE ID       CREATED        SIZE
nginx               alpine      812d47f806db   13 days ago    62.2MB
docker-postgres     latest      d700924fab98   4 months ago   453MB
find_your_team-web  latest      6612bb366874   5 months ago   486MB
<none>              <none>      fb93212c91c4   5 months ago   486MB
<none>              <none>      58d678653c4d   5 months ago   462MB
<none>              <none>      a855d30065a6   5 months ago   462MB
<none>              <none>      fd8958e69fe5   5 months ago   462MB
<none>              <none>      7b648ed4a205   5 months ago   462MB
postgres            15          f73372f4b458   5 months ago   444MB
dpage/pgadmin4      latest      f44751b4226c   5 months ago   540MB
<none>              <none>      226a962a7ee9   16 months ago  242MB
simpleuimasterr-web  latest      ee85fd96a281   16 months ago  242MB
```

postgres 15 уже установлен

```
PS D:\Dev\containerisation\containers-lab-1> docker history nginx:alpine
>>
IMAGE          CREATED          CREATED BY          SIZE      COMMENT
812d47f806db   13 days ago     RUN /bin/sh -c set -x && apkArch="$(cat _  49.4MB    buildkit.dockerfile.
v0
<missing>      13 days ago     ENV ACME_VERSION=0.3.1          0B        buildkit.dockerfile.
v0
<missing>      13 days ago     ENV NJS_RELEASE=1              0B        buildkit.dockerfile.
v0
<missing>      13 days ago     ENV NJS_VERSION=0.9.6          0B        buildkit.dockerfile.
v0
<missing>      13 days ago     CMD ["nginx" "-g" "daemon off;"] 0B        buildkit.dockerfile.
Finish Setup
```

11:52
29.04.2026

```
PS D:\Dev\containerisation\containers-lab-1> docker rmi nginx:alpine
Untagged: nginx:alpine
Untagged: nginx@sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de
Deleted: sha256:812d47f806db497c53f9b47e76bdab38bcf5d724c69d3986df5ee4336c210559
Deleted: sha256:27169268e599542a8efe40bc98eec72801e56c7eec6974fe72287429b5a56a88
Deleted: sha256:7e21891e773c24466c13d24a8f0fd9cdcc0310c88dfb9cfd5bd2a0480b48bd3e
Deleted: sha256:198fb1785a3693e2164fcfa5cccfed50c1ec233877f44a78189521a39a7c36e0
Deleted: sha256:cccb2672fed3544941cbd10edcd30161ce0abeec1ac6d02e9a7e758b2dd0eb50
Deleted: sha256:1d9a0e936fac6798e368a505eeb532babea0f78a88ccf678f7a7f6119d6af92a
Deleted: sha256:3f1f2c7545807c52541c2414015214c10174d63181a6a6dea381caf2386b23fe
Deleted: sha256:cbd8d5f3f17376aa3b25a5b2361749140b1ec2785fa186b3ad473675a341c218
Deleted: sha256:29df493baa13de438d6d2ece3a8333032e0b7b9b9d8cce4ee82194da255f61e1
PS D:\Dev\containerisation\containers-lab-1>
```

Finish Setup

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```
PS D:\Dev\containerisation> docker pull golang:1.21
1.21: Pulling from library/golang
903681d87777: Downloading 2.032MB/49.55MB
3cbbe86a28c2: Downloading 2.703MB/24.05MB
6ed93aa58a52: Downloading 2.703MB/64.14MB
1f46bd02dde3: Waiting
54bf7053e2d9: Waiting
532d43a0bc41: Waiting
4f4fb700ef54: Waiting
```

1.2 Работа с контейнерами

1.2 Жизненный цикл контейнеров

```
PS D:\Dev\containerisation> docker run -it --name test-alpine alpine:latest sh
latest: Pulling from library/alpine
6a0ac1617861: Already exists
Digest: sha256:5b10f432ef3da1b8d4c7eb6c487f2f5a8f096bc91145e68878dd4a5019afde11
Status: Downloaded newer image for alpine:latest
/ #
/ # ^C

/ # ls
bin      etc      lib      mnt      proc     run      srv      tmp      var
dev      home    media    opt      root     sbin     sys      usr
```

```
● PS D:\Dev\containerisation> docker run -d --name web-server -p 8080:80 nginx:alpine
47e09ce8a71b83bc8941cb9ef952f12672e8b7dd7f845b0ef639d7cc7813e2d2
● PS D:\Dev\containerisation> docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
	NAMES				
47e09ce8a71b	nginx:alpine	"/docker-entrypoint..."	16 seconds ago	Up 15 seconds	0.0.0
.0:8080->80/tcp	web-server				
157078e2c6f7	alpine:latest	"sh"	4 minutes ago	Exited (127) 3 minutes ago	
	test-alpine				

```
○ PS D:\Dev\containerisation>
```

Finish Setup

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```

PS D:\Dev\containerisation> docker exec -it web-server sh
/ # cat /usr/share/nginx/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit

```

What's next:

Finish Setup



14:50
29.04.2026

```

PS D:\Dev\containerisation> docker stop web-server
web-server
PS D:\Dev\containerisation> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS              PORTS
NAMES
47e09ce8a71b   nginx:alpine   "/docker-entrypoint...." 3 minutes ago  Exited (0) 3 seconds ago
web-server
157078e2c6f7   alpine:latest  "sh"                    8 minutes ago  Exited (127) 6 minutes ago
test-alpine
PS D:\Dev\containerisation> docker start web-server
web-server
PS D:\Dev\containerisation> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS              PORTS
NAMES
47e09ce8a71b   nginx:alpine   "/docker-entrypoint...." 4 minutes ago  Up 5 seconds        0.0.0.0:8080->80/tcp
web-server
157078e2c6f7   alpine:latest  "sh"                    8 minutes ago  Exited (127) 7 minutes ago
test-alpine
PS D:\Dev\containerisation> docker stop web-server
web-server
PS D:\Dev\containerisation> docker rm web-server
web-server
PS D:\Dev\containerisation> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS              PORTS      NAMES
157078e2c6f7   alpine:latest  "sh"                    9 minutes ago  Exited (127) 7 minutes ago
test-alpine
PS D:\Dev\containerisation>

```

Finish Setup



```
PS D:\Dev\containerisation> docker run -d --name postgres -e POSTGRES_PASSWORD=1234 postgres:15
2a9590e13ab656f1a34d3832d84f4dfab0d01f40eda01852e82be58ed7249ed7
PS D:\Dev\containerisation> docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
2a9590e13ab6   postgres:15 "docker-entrypoint.s..." 9 seconds ago  Up 8 seconds  5432/tcp     postgres
PS D:\Dev\containerisation> docker exec -it postgres sh
#
```

```
version
-----
PostgreSQL 15.15 (Debian 15.15-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2
.0, 64-bit
(1 row)
(END)
```

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1.3 Работа с томами

```

PS D:\Dev\containerisation> docker volume create my-app-data
my-app-data
PS D:\Dev\containerisation> docker volume ls
DRIVER      VOLUME NAME
local       0bfc282cdf25cfbc911ff9506af8c5a0cff94392a868e674b82148cfaf4c352c
local       72b9ec318610f0a60f1ade5ec07e689c8afe26676618a57724a196b6caea0c79
local       36492d0aaebd6179e93008b5f19b7dcd9f27ee908d8944eb89bc7cb8627fd123
local       a4531685224cec8d134a1b8dba415a99cc029888e9a5dc89e1ec04f846b686bd
local       d4705de2a78f516c27e06a56d5696e1ed9e6d2b266f3ad18487416ce875cc795
local       e522d10f8a93e1387744fcabbaf7a0f20054e9518b6a51261660ec5197e3d68
local       ed83e8c1120e5162926e07dd1c48633d28d563e111a46abdd7b977249355f7cc
local       f6f5acd24d9daf1e7d5320bcf6c0b32bbddb47178b8a446ffacdf2888a63cd0c
local       f803ff7cb792d4f31b66143dd1f1c37157f93d60e4b889dd1d7a22ca4bd4b479
local       ffa83283332d29dbf3508ad81306a4a1a9235babe5b0aeadc8c363fa724eb07
local       find_your_team_postgres_data
local       my-app-data
PS D:\Dev\containerisation> docker volume inspect my-app-data
[
  {
    "CreatedAt": "2026-04-29T12:06:07Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/my-app-data/_data",
    "Name": "my-app-data",
    "Options": null,
    "Scope": "local"
  }
]
PS D:\Dev\containerisation>

```

```

PS D:\Dev\containerisation> docker run -d `
>> --name postgres-db `
>> -e POSTGRES_PASSWORD=1234 `
>> -e POSTGRES_DB=testdb `
>> -v my-app-data:/var/lib/postgresql/data `
>> -p 5432:2345 postgres:15-alpine
c1b3701eb9ab65d896c341cca668bc3562c252ee2e76fd814b2a5dd7faf0b558
PS D:\Dev\containerisation> ps -a
Get-Process : Не удается найти параметр, соответствующий имени параметра "a".
строка:1 знак:4
+ ps -a
+ ~~~
+ CategoryInfo          : InvalidArgument: (:) [Get-Process], ParameterBindingException
+ FullyQualifiedErrorId : NamedParameterNotFound,Microsoft.PowerShell.Commands.GetProcessCommand

PS D:\Dev\containerisation> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
c1b3701eb9ab   postgres:15-alpine                 "docker-entrypoint.s..." 9 seconds ago  Up 8 seconds  5432/tcp, 0.0
.0.0:5432->2345/tcp   postgres-db
2a9590e13ab6   postgres:15                         "docker-entrypoint.s..." 21 minutes ago  Up 21 minutes  5432/tcp
postgres
PS D:\Dev\containerisation>

```

Finish Setup

15:21

29.04.2026


```

PS D:\Dev\containerisation> docker exec -it postgres-db psql -U postgres -d testdb -c "CREATE TABLE users (
id SERIAL, name TEXT);"
CREATE TABLE

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug post
gres-db
Learn more at https://docs.docker.com/go/debug-cli/
PS D:\Dev\containerisation> docker exec -it postgres-db psql -U postgres -d testdb -c "INSERT INTO users (n
ame) VALUES ('CHEGLAKOV_MAXIM_VITALEVICH BS80-16-23')"
INSERT 0 1

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug post
gres-db
Learn more at https://docs.docker.com/go/debug-cli/
PS D:\Dev\containerisation> docker stop postgres-db
postgres-db
PS D:\Dev\containerisation> docker rm postgres-db
postgres-db
PS D:\Dev\containerisation>

```

Finish Setup

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29.04.2026

```

PS D:\Dev\containerisation> docker stop postgres-db
postgres-db
PS D:\Dev\containerisation> docker rm postgres-db
postgres-db
PS D:\Dev\containerisation> docker run -d `
>> --name postgres-db `
>> -e POSTGRES_PASSWORD=1234 `
>> -e POSTGRES_DB=testdb `
>> -v my-app-data:/var/lib/postgresql/data `
>> -p 5432:2345 `postgres:15-alpine
67dfa1aac1829a1289f38e90112bd6271be159677b974fde20996a3835ddad07
PS D:\Dev\containerisation> docker exec -it postgres-db psql -U postgres -d testdb -c "SELECT * FROM users;"
 id |
-----+-----
  1 | CHEGLAKOV_MAXIM_VITALEVICH BS80-16-23
(1 row)

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug post
gres-db
Learn more at https://docs.docker.com/go/debug-cli/
PS D:\Dev\containerisation>

```

Finish Setup

15:27
29.04.2026

```

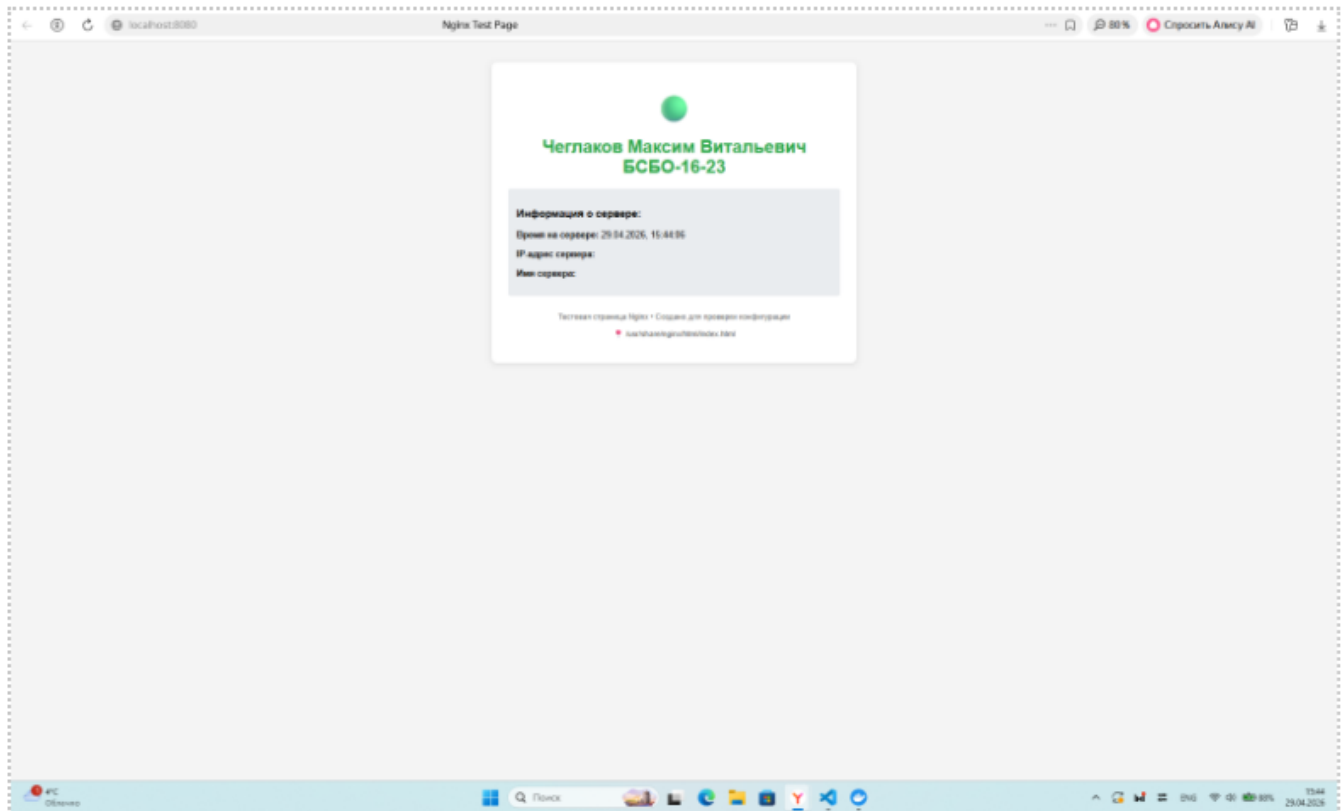
PS D:\Dev\containerisation> docker run -d `
>> --name nginx-cheglakov `
>> -p 8080:80 `
>> -v static-files:/usr/share/nginx/html `
>> nginx:alpine
05a0e1eb6508e906ad68d91f473bbbed3dd973f58eef2aa50d4035df1d51ae55c

```

```

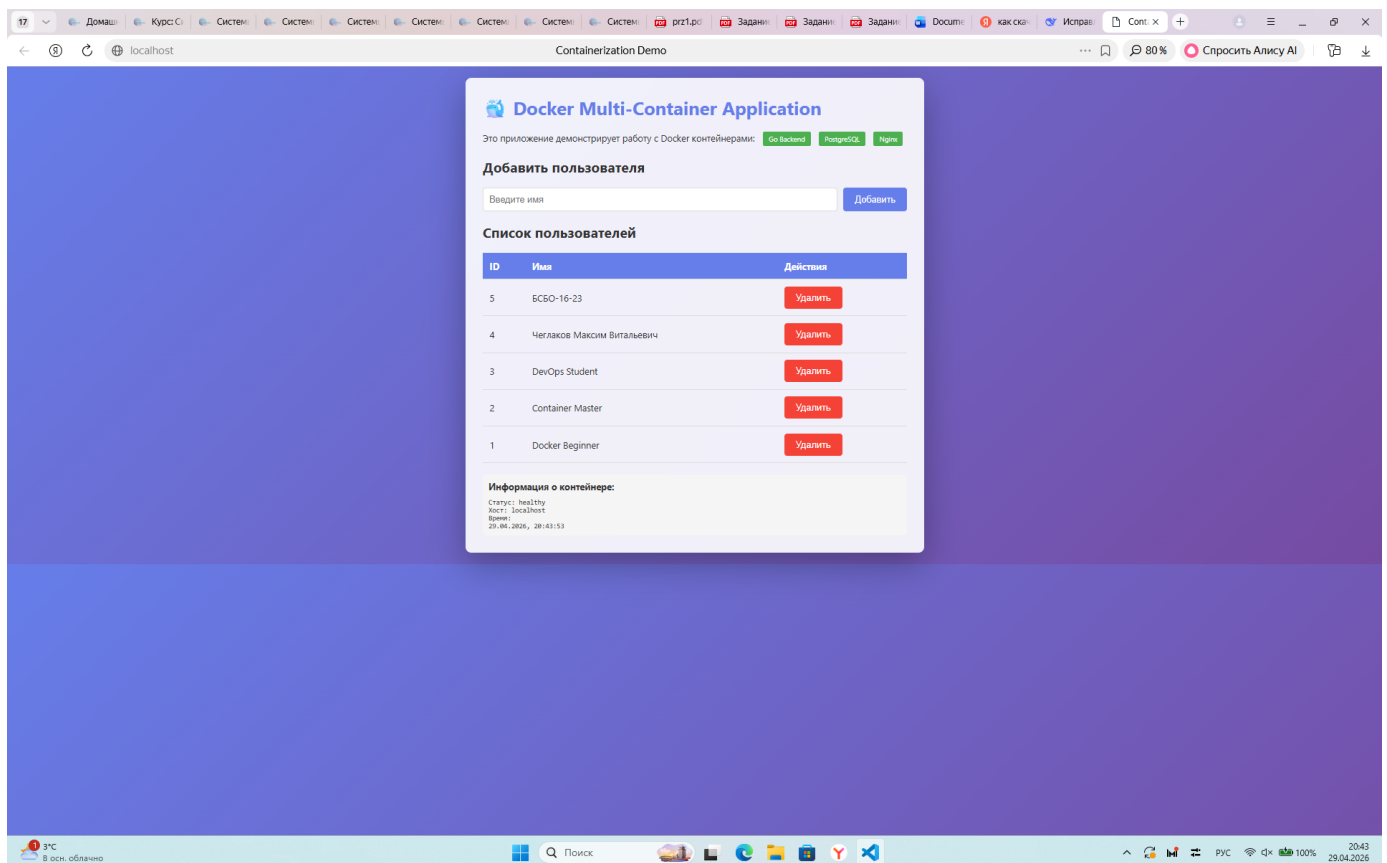
PS D:\Dev\containerisation\containers-lab-1\nginx> docker cp index.html nginx-cheglakov:/usr/share/nginx/ht
ml
Successfully copied 3.58kB to nginx-cheglakov:/usr/share/nginx/html
PS D:\Dev\containerisation\containers-lab-1\nginx>

```



2. Скриншоты работающего приложения

2.1 Главная страница



2.2 Список пользователей в БД

```
PS D:\Dev\containerisation\containers-lab-1> docker exec -it lab1_postgres psql -U postgres -d myapp -c "SELECT * FROM users;"
id |      name      |      created_at
---+-----+-----
 1 | Docker Beginner | 2026-04-29 15:58:29.009626
 2 | Container Master | 2026-04-29 15:58:29.009626
 3 | DevOps Student  | 2026-04-29 15:58:29.009626
 4 | Челлаков Максим Витальевич | 2026-04-29 17:43:12.702924
 5 | БСБО-16-23     | 2026-04-29 17:43:31.855004
(5 rows)
```

3. GitHub Actions

3.1 Успешный запуск workflow

← Build and Push Docker Images

✓ fix build 03 #16

Re-run all jobs

Summary

All jobs

build

Run details

Usage

Workflow file

Triggered via push 2 minutes ago

Maxim-Chegg pushed · e4b34d2 · main

Status

Success

Total duration

1m 44s

Artifacts

—

docker-build.yml

on: push

build 1m 34s

3.2 Опубликованные образы в GHCR

Package containers-lab-1/ Maxim-Chegg/container: x +

github.com

Maxim-Chegg/containers-lab-1

containers-lab-1 Public

main 1 Branch 1 Tag

Go to file

Add file

Code

MaximChegg trigger rebuild ✓ 56e74a4 · 6 minutes ago 15 Commits

.github/workflows

fix build 03

18 minutes ago

app

fix build 2

1 hour ago

init-scripts

my app

3 hours ago

nginx

fix build 2

1 hour ago

REPORT.md

my app

3 hours ago

docker-compose.test.yml

fix build 03

18 minutes ago

docker-compose.yml

fix build 01

41 minutes ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases 1

1 release latest 3 minutes ago

Packages 2

containers-lab-1/app

containers-lab-1/nginx

Contributors 1

MaximChegg Maxim Cheglakov

Languages

HTML 48.3%

Go 38.2%

Dockerfile 13.5%

4. Выводы

Узнал, как публиковать пакеты и настраивать сборку github actions

